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Personal Statement

About Me

I am a penultimate year computer science undergraduate at the National University of Singapore (NUS). Before studying at NUS, I was studying at Singapore Polytechnic where I was awarded with a Diploma in Infocomm Security Management. During the course of my studies, I was exposed to offensive cybersecurity attacks, defensive methods and investigative skills. Outside of school curriculum, I started to learn and develop web and mobile applications. I have also developed web and mobile (Android and iOS) apps and have deployed it into the respective mobile stores. When am I free, I would participate in hackathons as well as local tech meet-up events. I look forward to bringing not only my skills but also my general passion of using technology to help improve everyone's life, to your organization.

Internship Objectives

I would like to further develop my expertise in software engineering and gain more experience in developing production grade software.

Strengths & Weaknesses

My greatest strength is my ability to learn and adapt to changes quickly. Being able to learn fast has helped me to produce more results during my previous internships as the companies were using frameworks which I have not used before. There are times where I would be too detailed-oriented.

Work Experience

I have completed three software engineering/development internships. My first internship was at a local cyber security solutions company. I was tasked to develop a web app to visualize cyber-attacks for their customers. The visualization tool was created using D3JS and Vaadin. The web app also performed data analytics on cyber-attacks data which provided their staff with more insights, allowing them to respond faster to incidents.

During my second internship, I led a team of four members to assist a PhD student to enhance YALIH, a honeypot client. We improved the efficiency of it by developing a tool to easily analyse and generate reports of attack data which was extracted from the honeypot.

My third internship was at a defence agency which is responsible for the development of Singapore's defence systems. During this internship, I was tasked to digitalise the process of mobilizing military troops. This project required me to conduct user research, wireframing and prototyping. The final product was a web and mobile app, which was completed in a short time span of three months from start to end.

Skill Sets

My skill sets are not just limited to software engineering. I also have experience with system administrative tasks such as deploying applications/websites on VPS from Microsoft Azure and AWS.

How Can I Contribute

As a developer with five years of experience, I am experienced and comfortable working as a full-stack, frontend or backend engineer. I enjoy working on challenging issues and look forward to solving them. I am also a keen learner who is able to learn and adapt to new technologies quickly.

Education

Aug 2017 - Present	National University of Singapore Bachelor of Computing (Honours) in Computer Science (Course details in Appendix A)	Singapore
Aug 2015 - Aug 2017	Singapore Armed Forces, 1st Transport Battalion Transport Operator, Safety Coordinator - Compiled monthly/annually traffic accident safety reports - Transport military personnel to various military training locations	Singapore
Apr 2012 - Mar 2015	Singapore Polytechnic Diploma in Infocomm Security Management with Merit (Course details in Appendix B) - Cumulative GPA: 3.925 / 4.0 - Awarded Diploma with Merit (Top 10% of Cohort)	Singapore

Work Experience

May 2019 – Aug 2019	Defence Science Technology Agency Software Engineer Intern - Conducted user research, wireframing and prototyping - Developed a mobile and web app to track personnel in real-time - Utilised public-private key encryption to ensure data security	Singapore
Nov 2014 - Feb 2015	Victoria University of Wellington Software Developer Intern Led a team of three to develop a report generation tool for a honeypot client	Wellington, New Zealand
Sep 2014 – Nov 2014	e-Cop Software Developer Intern Developed a visualization web app using D3JS and Vaadin for company to identify cyber-attacks for their utility, rail, oil and gas industrial companies customers	Singapore

Scholastic Achievements/Extracurricular Activities

Apr 2012 - Present	Integrated Infocomm Scholarship Info-communications Media Development Authority of Singapore scholarship	Singapore
Dec 2018 – Jan 2019	AIESEC Global Volunteer Program Education Volunteer - Planned and conducted web development classes for 10 students - Enhanced local committee organisation's website	Agadir, Morocco
Sep 2014	The Software Developer Examination – Java Proxor 3 (Certification Level)	Singapore
Dec 2013	Youth Expedition Project Education Volunteer - Shared photoshop and video editing skills with local students - Collaborated and organised a fund-raising event	Ho Chi Minh, Vietnam

Skill Sets & Proficiency

Programming	Proficient: Java, Python Intermediate: Swift, JavaScript Basic: C
Web/Mobile Development	Proficient: HTML, CSS, JSP, Bootstrap Intermediate: Django, Flask, Spring, Cloudflare, Android, iOS Basic: PHP

Database	Proficient: MySQL DBMS, Microsoft SQL Intermediate: DynamoDB
Version Control	Intermediate: Git, GitHub
DevOps	Intermediate: Apache, Heroku, Microsoft Azure, TravisCI, JUnit Basic: AWS Elastic Beanstalk
Multimedia	Intermediate: Adobe Photoshop CS6, Adobe Premiere Pro CS6

Language Proficiency

Spoken	English – fluent; Mandarin – fluent
Written	English – competent; Chinese – average

Programming Related Projects

Feb 2019	4 Player Pong <i>Developer</i> • Tech Stack: WebRTC, JavaScript, HTML & CSS • Developed a multiplayer web app game which uses mobile phones as controllers and a shared display within 24 hours at a hackathon • Implemented the game control connection between mobile phones and the game
Aug 2018 – Nov 2018	modsUni <i>Developer</i> • Tech Stack: Java, JavaFX, Junit • Project Portfolio: https://bit.ly/ReubenModsUniPortfolio • Developed a Java desktop app for students to plan and manage modules to be taken throughout their university education • Implemented “Generate Schedule” features which includes an algorithm to check for pre-requisite conditions • Wrote unit test cases for the algorithm and feature
Aug 2018	Carpark SG <i>Sole Developer</i> • Tech Stack: Java, Swift, Cloudflare, Spring, Land Transport Authority APIs • Available at: http://carparksg.tk • Built native iOS app that helps drivers to find parking lots using real-time lot availability information • Implemented a restful API service which utilized Cloudflare’s CDN for caching and to reduce latency
May 2018 - Aug 2018	Articully <i>Lead Developer</i> • Tech Stack: Python, Django, Microsoft Azure, AWS DynamoDB, Cloudflare, Braintree Payment • Collaborated with another member to build an ecommerce platform that connects freelance writers and publishers together • Implemented search, user review system, login/logout, payment and 2FA features
Jun 2017	Individual Physical Proficiency Test (IPPT) Calculator <i>Sole Developer</i> • Tech Stack: Java • Developed a native Android app that helps uniformed servicemen/women who are in Singapore’s military, police or civil defence force to determine their IPPT standards (Pass, Silver, Gold)
Apr 2014 – Aug 2014	Open Bouncer <i>Lead Developer</i> • Tech Stack: Python, Flask, Django, Android Emulators, Linux VPS • Featured on Singapore Hometeam website: https://bit.ly/ReubenOpenBouncer • Developed a framework for detecting malicious Android applications • Implemented web frontend, restful API service and dynamic Android app analysis

Degree: Bachelor of Computing (Honours) in Computer Science
Cumulative Average Point: 3.87 / 5.00

Category	Course Description	Grades
Programming	Programming Methodology II	B+
	Data Structures and Algorithms	A
Development	Software Engineering	A-
	Interaction Design	B
	Independent Software Development Project (Orbital)	CS
	Formal Methods for Software Engineering	IP
	Software Engineering and Principles	IP
Computer Network	Introduction to Computer Networks	B+
	Introduction to Operating Systems	B+
	Internet Architecture	IP
Computer Architecture	Computer Organisation	B
Communications	Effective Communication for Computing Professionals	B+
Mathematics	Introductory Mathematics	A-
	Linear Algebra I	B
	Calculus for Computing	S
	Probability and Statistics	C+
	Discrete Structures	S
Electives	Public Health in Action	A
	Natural Heritage of Singapore	B+
	Quantitative Reasoning	B
	Operations and Technology Management	S
	Asking Questions	CS
	Sky and Telescopes	IP
	Introduction to Artificial Intelligence	C+
	Computer Security	W

NUS Grading Scale:

A+ & A (5.0); A- (4.5); B+ (4.0); B (3.5); B- (3.0); C+ (2.5); C (2.0); D+ (1.5); D (1.0); F (0)

S = Satisfactory; U = Unsatisfactory

CS = Completed Satisfactorily; CU = Completed Unsatisfactorily

EXE = Exempted; IC = Incomplete; IP = In Progress; W = Withdrawn

Diploma: Diploma in Infocomm Security Management with Merit
Cumulative Grade Point Average: 3.925 / 4.000

Category	Course Description	Grades
Programming	Java Programming	DIST
	Advanced Java Programming	DIST
	Programming Using PERL and C	A
Database	Database Management Systems	DIST
	Database Security and Management	A
Development	Web Client Development	B+
	Enterprise Application Development	DIST
	Infosec Project Development and Management	A
Cyber Security	Applied Cryptography	A
	Ethical Hacking and Defences	A
	Infocomm Security	A
	Computer Law & Investigation	A
	Computer Forensics	A
Sysadmin	Network and Operating Systems	A
	Essential Linux System Administration	A
	Securing Linux	A
	Securing Microsoft Windows	A
	IT Security Policies and Auditing	A
	Operations Security and Contingency Management	DIST
Communications	Teamwork and Communication Skills	DIST
	iDiscover	DIST
	Social Innovation Project	A
	Workplace Communication Skills	A
Electives	General Education 1	A
	General Education 2	A
	General Education 3	B+
	Mathematics	B
	Digital Visual Design	A
	Social Media Marketing	B+

Singapore Polytechnic Grading Scale:

DIST (4.0); A (4.0); B+ (3.5); B (3.0); C+ (2.5); C (2.0); D+ (1.5); D (1.0); D- (0.5); F (0)